A Collection of Proofs

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1 Discrete Mathematics

1.1 Logic, Sets, Quantifiers

1. **Cardinality of Power Sets.** If A is a finite set with n elements, then $|\mathscr{P}(A)| = 2^n$.
2. **Commutability of Nested Quantifiers.** If $\exists x\ \forall y\ P(x, y)$, then $\forall y\ \exists x\ P(x, y)$. However, the converse is not necessarily true.
3. **Functional Completeness of $\{\neg, \wedge\}$.** The set of logical connectives $\{\neg, \wedge\}$ is functionally complete. That is, every Boolean function $\{0, 1\}^n \to \{0, 1\}$ can be expressed using only the connectives $\neg$ and $\wedge$.
4. **Russell's Paradox.** For any set A, the set $\{x \in A \mid x \notin x\}$ is not an element of A.

1.2 Functions, Inverses, Composition

1. Let $f : A \to B$ and $g : B \to C$ be functions. The following are true:
   a. If f and g are both injective, then the composition $g \circ f$ is also injective.
   b. If f and g are both surjective, then the composition $g \circ f$ is also surjective.
   c. If $g \circ f$ is injective, then f is injective, but not necessarily g.
   d. If $g \circ f$ is surjective, then g is surjective, but not necessarily f.
2. **Characteristic Function.** For any set $A \subseteq U$, define its characteristic function $\chi_A : U \to \{0, 1\}$ by:

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

   The following are true:
   a. $\chi_{A\cap B} = \chi_A \chi_B$.
   b. $\chi_{A\cup B} = \chi_A + \chi_B - \chi_A \chi_B$.
   c. $\chi_{A^\complement} = 1 - \chi_A$.
   d. The map $A \mapsto \chi_A$ is a bijection from $\mathscr{P}(U)$ and the set of functions $U \to \{0, 1\}$.

1.3 Induction, Well-Ordering

1. **Bernoulli's Inequality.** For any real number $x \geq -1$ and any integer $n \geq 0$, we have:

$$(1 + x)^n \geq 1 + nx.$$

2. **Fundamental Theorem of Arithmetic.** Every integer $n \geq 2$ has a unique prime factorization.
3. **AM-GM Inequality.** For any nonnegative real numbers $a_1, a_2, \ldots, a_n$, we have:

$$\left(a_1 a_2 \ldots a_n\right)^{1/n} \leq \frac{a_1 + a_2 + \ldots + a_n}{n}.$$

4. **Towers of Hanoi.** For any integer $n \geq 1$, the minimum number of moves required to solve the Towers of Hanoi puzzle with n disks is $2^n - 1$.

1.4 Relations, Equivalence Relations, Partitions

1.5 Finite, Countable, Uncountable Sets

1.6 Number Theory

1.7 Combinatorics, Probability

1. **Formula for Combination.** For any integers $n \geq 0$ and $0 \leq k \leq n$, the number of ways to choose k elements from a set of n distinct elements is given by the formula:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

2. **Pascal's Identity.** For any integers $n \geq 1$ and $0 < k \leq n$:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

3. **Binomial Theorem.** For any numbers a and b, and integer $n \geq 0$:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

4. **Inclusion-Exclusion Principle.** For any finite sets $A_1, A_2, \ldots, A_n$:

$$\left|A_1 \cup A_2 \cup \ldots \cup A_n\right| = \sum_{i=1}^{n} \left|A_i\right| - \sum_{1 \leq i < j \leq n} \left|A_i \cap A_j\right| + \ldots + (-1)^{n+1} \left|A_1 \cap A_2 \cap \ldots \cap A_n\right|.$$

5. **Bayes' Theorem.** Let A and B be events with $P(B) > 0$. Then:

$$P(A|B) = \frac{P(B|A)\, P(A)}{P(B)}.$$

1.8 Real Numbers, Inequalities, Convergence

1.9 Orders, Lattices

1.10 Recursion, Algorithms

1.11 Graph Theory

1. **Handshake Lemma.** Let $G = (V, E)$ be a finite graph. Then the total degree of the vertices in G is twice the number of edges in G, i.e.

$$\sum_{v \in V} \deg(v) = 2|E|.$$

2. A graph is bipartite **iff** it contains no odd cycles.

2 Linear Algebra

2.1 Vector Spaces, Subspaces, Linear Independence

Let V be a vector space.

1. Let $V_1, V_2, \dots, V_p$ be subspaces of V. The intersection $V_1 \cap V_2 \cap \dots \cap V_p$ is a subspace of V.
2. Let V_1 and V_2 be subspaces of V. The union $V_1 \cup V_2$ is a subspace **iff** $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.
3. Let $S \subseteq V$. Then $\operatorname{span}(S)$ is the smallest subspace of V containing S. That is, $S \subseteq \operatorname{span}(S)$, and $\forall$ subspaces W of V, $W \supseteq S \Rightarrow \operatorname{span}(S) \subseteq W$.
4. A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ with $n \geq 2$ is linearly dependent **iff** $\exists \mathbf{v}_i \in S$ such that $\mathbf{v}_i$ is a linear combination of the other vectors in S.
5. If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent but $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{w}\}$ is linearly dependent, then $\mathbf{w} \in \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.
6. **Telescoping Linear Independence.** Suppose $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set of vectors in V. Define a new set of vectors $W = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ by $\mathbf{w}_i = \sum_{j=1}^{i} \mathbf{v}_j$ for $i = 1, 2, \dots, n$, so $W = \{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \dots, \mathbf{v}_1 + \mathbf{v}_2 + \dots + \mathbf{v}_n\}$. Then W is also a linearly independent set.
7. **Steinitz Exchange Lemma.** Let $U = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \subseteq V$ and $W = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\} \subseteq V$. If U is linearly independent and W spans V, then $m \leq n$, and we can replace m vectors from W with the m vectors from U to get a spanning set for V.
8. **Basis Theorem.** If V has dimension n, then:
   a. If $\mathcal{B}$ is a linearly independent set of n vectors from V, then $\mathcal{B}$ is a basis for V.
   b. If $\mathcal{B}$ is a set of n vectors from V that span V, then $\mathcal{B}$ is a basis for V.
9. If S spans V, then a basis $\mathcal{B}$ for V can be formed using only vectors in S, i.e. $\mathcal{B} \subseteq S$.
10. For any two subspaces V_1 and V_2 of V, $\dim(V_1 + V_2) = \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2)$.
11. **Invariant Intersections.** Suppose V is finite-dimensional, and U and W are subspaces of V such that $\dim U + \dim W > \dim V$. Then $U \cap W$ must contain a nonzero vector.
12. Let $a_1, a_2, \dots, a_n$ be distinct scalars in $\mathbb{R}$. Then the set of functions $\{e^{a_1 x}, e^{a_2 x}, \dots, e^{a_n x}\}$ is linearly independent in the set of continuous functions $C(\mathbb{R})$.
13. $\mathbb{R}^\infty$ is the infinite-dimensional vector space containing all infinite sequences of real numbers $(a_1, a_2, a_3, \dots)$. It holds that $\{(1, b, b^2, b^3, \dots) \in \mathbb{R}^\infty \mid b \in \mathbb{R}\}$ is a linearly independent set in $\mathbb{R}^\infty$.

Solutions

2.2 Linear Maps, Kernel, Image, Rank-Nullity

Note: A linear map is also called a linear transformation. The image of a linear map T, denoted $\mathrm{im}(T)$, is sometimes called the range of T.

1. **Rank-Nullity Theorem for Matrices.** For any $m \times n$ matrix A, $\mathrm{rank}(A) + \mathrm{nullity}(A) = n$.
2. **Rank-Nullity Theorem for Linear Maps.** Let $T : V \to W$ be a linear map, where V is finite-dimensional. Then:
$$\dim(\ker T) + \dim(\mathrm{im}\ T) = \dim(V).$$
3. **Equivalence of Column and Row Ranks.** If A is an $m \times n$ matrix, then:
$$\dim(\mathrm{col}\ A) = \dim(\mathrm{row}\ A) = \mathrm{rank}(A).$$
4. **Sylvester's Law of Nullity.** For linear maps $T : U \to V$ and $S : V \to W$, then:
$$\mathrm{nullity}(S \circ T) \leq \mathrm{nullity}(S) + \mathrm{nullity}(T).$$
5. **Sylvester's Rank Inequality.** For linear maps $T : U \to V$ and $S : V \to W$, then:
$$\mathrm{rank}(S \circ T) \geq \mathrm{rank}(S) + \mathrm{rank}(T) - \dim(V).$$
6. **Frobenius Rank Inequality.** For matrices A, B, C of compatible dimensions:
$$\mathrm{rank}(AB) + \mathrm{rank}(BC) \leq \mathrm{rank}(B) + \mathrm{rank}(ABC).$$
7. **Rank of Compositions.** For linear maps $T : U \to V$ and $S : V \to W$, then:
$$\mathrm{rank}(T \circ S) \leq \min(\mathrm{rank}(T), \mathrm{rank}(S)).$$
8. **Existence and Uniqueness of Linear Maps.**
   a. **Uniqueness.** Suppose $T, S : V \to W$ are linear maps, and $\mathcal{B} = \left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\right\}$ is a basis of V such that $T\left(\mathbf{v}_i\right) = S\left(\mathbf{v}_i\right)$ for $i = 1, 2, \ldots, n$. Then $T = S$.
   b. **Existence.** Let $\mathcal{B} = \left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\right\}$ be a basis of V, and let $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n$ be any vectors in W. Then there exists a linear map $T : V \to W$ such that $T\left(\mathbf{v}_i\right) = \mathbf{w}_i$ for $i = 1, 2, \ldots, n$.
9. Let $T : V \to W$ be linear. Then:
   a. $\ker(T)$ is a subspace of V and $\mathrm{im}(T)$ is a subspace of W.
   b. Then T is injective **iff** $\ker(T) = \{\mathbf{0}\}$.
   c. Then T is surjective **iff** $\mathrm{im}(T) = W$.
10. Let V and W be finite-dimensional vector spaces over the same field $\mathbb{F}$. Then V and W are isomorphic **iff** $\dim(V) = \dim(W)$.
11. **Adjoints of Linear Maps.** Let $T : V \to V$ be a linear map on a finite-dimensional inner product space V, and T^* is the adjoint of T, i.e. $\langle \mathbf{u}, T(\mathbf{v})\rangle = \langle T^*(\mathbf{u}), \mathbf{v}\rangle$ for all $\mathbf{u}, \mathbf{v} \in V$. Show that:
   a. T^* exists and is unique.
   b. $\left(T^*\right)^* = T$.
   c. $(ST)^* = T^* S^*$ for another linear map $S : V \to V$.
   d. $\ker\left(T^*\right) = (\mathrm{im}\, T)^\perp$.
   e. T is self-adjoint ($T^* = T$) **iff** the associated matrix of T is a Hermitian matrix.

12. Let $L, R : \mathbb{R}^\infty \to \mathbb{R}^\infty$ be the left and right shift operators on $\mathbb{R}^\infty$, respectively, defined by $L\big(a_1, a_2, a_3, \ldots\big) = \big(a_2, a_3, a_4, \ldots\big)$ and $R\big(a_1, a_2, a_3, \ldots\big) = \big(0, a_1, a_2, a_3, \ldots\big)$. Then $L \circ R = \mathrm{id}_{\mathbb{R}^\infty}$, but $R \circ L \neq \mathrm{id}_{\mathbb{R}^\infty}$, where $\mathrm{id}_{\mathbb{R}^\infty}$ is the identity operator on $\mathbb{R}^\infty$ ($\mathbf{v} \mapsto \mathbf{v}$).
13. Let $T : V \to V$ be a linear map where V is finite-dimensional. If $\mathrm{im}(T) = \mathrm{im}(T \circ T)$, then $\ker(T) \cap \mathrm{im}(T) = \{\mathbf{0}\}$.
14. Let $T : V \to W$ be a linear map. If T is an isomorphism and $\big\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\big\}$ is a basis for V, then $\big\{T\big(\mathbf{v}_1\big), T\big(\mathbf{v}_2\big), \ldots, T\big(\mathbf{v}_n\big)\big\}$ is a basis for W.

2.3 Matrix Operations, Inverses, Determinants

1. **Existence of Row Echelon Form.** For any $m \times n$ matrix A, there must exist a row-equivalent matrix B in row echelon form.
2. **Uniqueness and Existence of Reduced Row Echelon Form.** For any $m \times n$ matrix A, there is a unique row-equivalent matrix B in reduced row echelon form.
3. For $n \times n$ matrices A and B, $\det(AB) = \det(A)\det(B)$.
4. If A is invertible, then $\det\left(A^{-1}\right) = \dfrac{1}{\det(A)}$.
5. If A is an $n \times n$ real matrix, then for some $k \in \mathbb{R}$, $\det(kA) = k^n \det(A)$.
6. If A is an $n \times n$ matrix such that $A^2 = A$, then $\det(A) \in \{0, 1\}$.
7. If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A.
8. If A is invertible, then $A^\top A$ is also invertible, and $A^{-1} = \left(A^\top A\right)^{-1} A^\top$.
9. Suppose A and B are matrices such that the product AB is defined and A and AB are both invertible. Then B is also invertible.
10. Let A be an $n \times n$ matrix. If $A^2 = 0_{n\times n}$ where $0_{n\times n}$ is the $n \times n$ zero matrix, then A is singular.
11. **Cramer's Rule.** Let A be an invertible $n \times n$ matrix, and let $\mathbf{b}$ be a vector in $\mathbb{R}^n$. Then the unique solution to the equation $A\mathbf{x} = \mathbf{b}$ is given by:

    $$x_i = \frac{\det\left(A_i\right)}{\det(A)} \text{ for } i = 1, 2, \ldots, n.$$

    where A_i is the matrix obtained from A by replacing the i^{th} column of A with the vector $\mathbf{b}$.
12. **Determinant of a Block Matrix.** Let M be the following block matrix:

    $$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

    where A and D are square matrices, and A is invertible. Then $\det(M) = \det(A)\det\left(D - CA^{-1}B\right)$.
13. **Invertible Matrix Theorem.** Let A be an $n \times n$ matrix. Then the following are equivalent:
    a. A is invertible.
    b. For some $\mathbf{x} \in \mathbb{R}^n$, if $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = \mathbf{0}$.
    c. $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ for every $\mathbf{b} \in \mathbb{R}^n$.
    d. The columns of A form a basis for $\mathbb{R}^n$.
    e. $\operatorname{rank} A = n$.
    f. $\det A \neq 0$.
    g. $\operatorname{nul} A = \{\mathbf{0}\}$.
    h. $A^\top$ is invertible.
    i. 0 is not an eigenvalue of A.
    j. A is row equivalent to the $n \times n$ identity matrix I_n.
    k. The linear map $\mathbf{x} \mapsto A\mathbf{x}$ is an isomorphism from $\mathbb{R}^n$ to $\mathbb{R}^n$.
    l. $A^\top A$ is positive definite.

14. **Convergence of Matrix Exponential.** Let A be any $n \times n$ complex matrix. Then the Taylor expansion of $e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$ converges.
15. **Determinant of Matrix Exponential.** Let A be any $n \times n$ complex matrix. Then $\det\left(e^A\right) = e^{\mathrm{tr}(A)}$.

2.4 Eigenvalues, Eigenvectors, Diagonalization

1. **Eigenvectors Corresponding to Distinct Eigenvalues are Linearly Independent.** Let $\lambda_1, \lambda_2, \ldots, \lambda_p$ be distinct eigenvalues of a matrix A, and let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be their corresponding eigenvectors. Then $\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\right\}$ is a linearly independent set.
2. **Diagonalization Theorem.** Let A be an $n \times n$ matrix. Then the following are equivalent:
   a. A is diagonalizable.
   b. There exists a basis for $\mathbb{R}^n$ consisting of eigenvectors from A.
   c. The sum of the dimensions of the eigenspaces of A is n.
   d. For each eigenvalue λ of A, the geometric multiplicity of λ equals the algebraic multiplicity of λ.
3. Let λ be an eigenvalue of an $n \times n$ matrix A. Let m_g be the geometric multiplicity of λ, and let m_a be the algebraic multiplicity of λ. Then $1 \leq m_g \leq m_a \leq n$.
4. If A and B are similar matrices, then:
   a. A and B have the same characteristic polynomial.
   b. $\det A = \det B$.
   c. $\operatorname{tr} A = \operatorname{tr} B$.
5. **Cayley-Hamilton Theorem.** Let A be an $n \times n$ matrix, and let $p(\lambda) = \det\left(\lambda I_n - A\right)$ be the characteristic polynomial of A. Then $p(A) = 0_{n\times n}$, where $0_{n\times n}$ is the $n \times n$ zero matrix.
6. Let A be a real matrix, and let $\lambda = a + bi$ ($b \neq 0$) be a complex eigenvalue of A. If $\mathbf{v} = \mathbf{u} + i\mathbf{w}$ is a corresponding eigenvector for λ, then:
   a. The complex conjugate $\lambda^* = a - bi$ is also an eigenvalue of A, with corresponding eigenvector $\mathbf{v}^* = \mathbf{u} - i\mathbf{w}$.
   b. A is similar to a block-diagonal matrix containing the block $C = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, where C is a rotation-scaling matrix in the basis $\{\mathbf{u}, \mathbf{w}\}$.
7. Let A have eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, counting algebraic multiplicities. Then:
   a. $$\det A = \lambda_1 \lambda_2 \ldots \lambda_n = \prod_{i=1}^{n} \lambda_i.$$
   b. $$\operatorname{tr} A = \lambda_1 + \lambda_2 + \ldots + \lambda_n = \sum_{i=1}^{n} \lambda_i.$$
8. If all eigenvalues of A have an absolute value less than 1, then $I - A$ is invertible.
9. If λ is an eigenvalue of an $n \times n$ matrix A, then for all integers $k \geq 1$, λ^k is an eigenvalue of A^k.
10. If A is a triangular matrix, then all eigenvalues of A are on the main diagonal of A.
11. If A is a 2×2 matrix with repeated eigenvalue λ, then A is diagonalizable **iff** $A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.
12. If A and B are $n \times n$ matrices with the same n distinct eigenvalues, then A is similar to B.

2.5 Inner Product Spaces, Orthogonality

Let V be an inner product space.

1. An orthogonal set of nonzero vectors is linearly independent.
2. **Completeness of Orthogonal Complements.** Let $W \subseteq V$ be finite-dimensional. Then:
   a. $\left(W^{\perp}\right)^{\perp} = W$.
   b. $\dim(W) + \dim\left(W^{\perp}\right) = \dim(V)$.
3. **Triangle Inequality.** Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in V. Then $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
4. **Cauchy-Schwarz Inequality.** Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in V. Then $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$.
5. **Bessel's Inequality.** Let $\mathcal{B} = \left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\right\}$ be an orthonormal set of vectors in V, and let $\mathbf{v}$ be any vector in V. Then:

$$\sum_{i=1}^{n} \left|\langle \mathbf{v}, \mathbf{v}_i \rangle\right|^2 \leq \|\mathbf{v}\|^2.$$

6. **Best Approximation Theorem.** Let W be a finite-dimensional subspace of V, and let $\mathbf{v} \in V$. Then for all $\mathbf{w} \in W$ such that $\mathbf{w} \neq \operatorname{proj}_W(\mathbf{v})$:

$$\left\|\mathbf{v} - \operatorname{proj}_W(\mathbf{v})\right\| < \|\mathbf{v} - \mathbf{w}\|.$$

7. **Eigenvalues of an Orthogonal Matrix.** If Q is an orthogonal matrix, then every eigenvalue of Q has an absolute value of 1.
8. **Fundamental Theorem of Linear Algebra.** Let A be an $m \times n$ matrix. Then:
   a. $\operatorname{nul}(A) = (\operatorname{row} A)^{\perp}$ in the auxiliary inner product space $\mathbb{R}^n$.
   b. $\operatorname{nul}\left(A^{\mathsf{T}}\right) = (\operatorname{col} A)^{\perp}$ in the auxiliary inner product space $\mathbb{R}^m$.
9. **Existence and Uniqueness of Orthogonal Decomposition.** Let W be a finite-dimensional subspace of V. Then, for any vector $\mathbf{v} \in V$, there is a unique $\hat{\mathbf{v}} \in W$ and $\mathbf{z} \in W^{\perp}$ such that $\mathbf{v} = \hat{\mathbf{v}} + \mathbf{z}$.
10. Let Q be an $n \times n$ real matrix. Then the following are equivalent:
    a. Q is orthogonal, i.e. $Q^{\mathsf{T}} = Q^{-1}$.
    b. The columns of Q form an orthonormal set.
    c. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle Q\mathbf{u}, Q\mathbf{v} \rangle$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. (preserves inner products)
    d. $\|Q\mathbf{v}\| = \|\mathbf{v}\|$ for all $\mathbf{v} \in \mathbb{R}^n$. (preserves norms)
11. **Normal Equation.** Let A be an $m \times n$ matrix, and let $\mathbf{b}$ be a vector in $\mathbb{R}^m$. Then the least-squares solutions to the equation $A\mathbf{x} = \mathbf{b}$ are the solutions to the normal equation $A^{\mathsf{T}}A\hat{\mathbf{x}} = A^{\mathsf{T}}\mathbf{b}$.

2.6 Symmetric & Complex Matrices, Quadratic Forms

1. **Spectral Theorem.** Let A be a real symmetric matrix. Then there exists an orthogonal matrix Q and a diagonal matrix D such that $A = QDQ^{\mathsf{T}}$.
2. If A is a real symmetric matrix, then A only has real eigenvalues.
3. Let A be an $n \times n$ complex matrix. Then the following are equivalent:
   a. A is normal, i.e. $A^{\dagger}A = AA^{\dagger}$.

b. A is unitarily diagonalizable.
c. A basis for $\mathbb{C}^n$ can be formed from eigenvectors of A.

2.7 Applications of Linear Algebra

3 Vector Analysis